

Homework #4:

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CHE 321

2/19/2020

1) a) $C_{A0} = C_{B0} = 2M$

$$K_c = 10 \frac{L}{mol}$$

$$K_c = \frac{C_c}{C_A C_B}$$

$$K_c = \frac{C_{A0} x}{C_{A0}^2 (1-x)^2}$$

$$C_{A0}^2 K_c (x^2 - 2x + 1) = C_{A0} x$$

$$C_{A0} K_c (x^2 - 2x + 1) = x$$

$$2 \frac{mol}{L} (10 \frac{L}{mol}) (x^2 - 2x + 1) = x$$

$$20(x^2 - 2x + 1) = x$$

$$20x^2 - 40x + 20 = x$$

$$20x^2 - 41x + 20 = 0$$

$$x = 1.25 \text{ } \frac{1}{2} x = 0.8$$

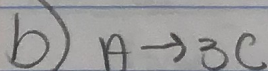
x should be less than one so

$$x = 0.8$$

$$C_A = C_{A0}(1-x) = 2(1-0.8) = 0.4 \text{ mol/L}$$

$$C_B = C_{A0}(1-x) = 2(1-0.8) = 0.4 \text{ mol/L}$$

$$C_c = C_{A0}(x) = 2(0.8) = 1.6 \text{ mol/L}$$



$$V_t = V_0(1-x) + 3V_0x$$

$$PV = nRT$$

$$C = \frac{n}{V}$$

$$P = CRT$$

$$K_c = \frac{C_c^3}{C_A}$$

$$C_A = \frac{n_A}{V} = \frac{N_A}{V} = \frac{N_{A0}(1-x)}{V_0(1+2x)} = C_{A0} \frac{(1-x)}{(1+2x)}$$

$$C_c = \frac{N_{A0}x}{V_0(1+2x)} = C_{A0} \frac{x}{1+2x}$$

$$K_c = 27 C_{A0}^3 x^3 \frac{(1+2x)}{(C_{A0}(1-x))^2} \Rightarrow \text{find } x$$

$$K_c = 27 C_{A0}^3 x^3$$

$$(1+2x)^2 C_{A0} (1-x)$$

$$\begin{matrix} 1+2x(1+2x) \\ 1+4x+4x^2 \end{matrix}$$

$$0.25 \left(\frac{L}{mol} \right)^2 = 27 (0.3047)^3 x^3$$

$$(1+2x)^2 0.3047 (1-x)$$

$$0.25 (1+2x)^2 0.3047 (1-x) = 27 (0.3047)^3 x^3$$

$$0.25 (1+2x)^2 0.3047 (1-x) = 0.7638 x^3$$

$$0.25 (1+4x+4x^2) 0.3047 - 0.3047 x = 0.7638 x^3$$

$$0.25 + x + x^2 (0.3047 - 0.3047 x) = 0.7638 x^3$$

$$0.076175 + 0.3047x + 0.3047x^2 - 0.076175x - 0.3047x^2 - 0.3047x^3 = 0.7638x^3$$

$$0 = 1.0685x^3 - 0.228525x - 0.076175$$

$$x = 0.58028$$

$$C_A = C_{A0} \frac{(1-x)}{(1+2x)}$$

$$= 0.3047 \frac{(1-0.58028)}{(1+2(0.58028))}$$

$$C_A = 0.0592 \text{ mol/L}$$

$$C_C = 3 C_{A0} \frac{x}{1+2x}$$

$$= 0.3047 \left(\frac{0.58028}{1+2(0.58028)} \right)$$

$$C_C = 0.2416 \text{ mol/L}$$

$$P_0 V_0 = n R T$$

$$\frac{P_0}{R T} = \frac{n_0}{V_0} = C_{A0}$$

$$C_{A0} = 1.0 \text{ atm}$$

$$(0.08206 \frac{\text{atm} \cdot \text{L}}{\text{mol} \cdot \text{K}}) (400 \text{ K})$$

$$C_{A0} = 0.3047 \frac{\text{mol}}{\text{L}}$$

$$2) n = \frac{V_{tot}}{V_{tube}}$$

$$V_{tube} = \pi r^2 L$$

$$PFR = \frac{dF_A}{dV} = -r_A$$

$$r_A = -k C_A = -k C_{A0}(1-x)$$

$$V_{tot} = F_{A0} \int_0^{0.9} \frac{dx}{-r_A} \Rightarrow F_{A0} \int_0^{0.9} \frac{dx}{k C_{A0}(1-x)} \Rightarrow \frac{F_{A0}}{k_2 C_{A0}} \int_0^{0.9} \frac{dx}{(1-x)}$$

$$V = \frac{F_{A0} \ln 10}{k C_{A0}} \quad \therefore \frac{k_2}{k_1} = e^{\frac{E}{R}(\frac{1}{T_2} - \frac{1}{T_1})} \Rightarrow k_2 = k_1 e^{\frac{E}{R}(\frac{1}{T_2} - \frac{1}{T_1})}$$

$$T_1 = 80^\circ F = 300K, T_2 = 260^\circ F = 400K$$

$$E_a = 25,000 \frac{\text{cal}}{\text{gmol}}$$

$$R = 1.9858775 \frac{\text{cal}}{\text{gmol} \cdot K}$$

$$k_2 = 0.0015 \text{ min}^{-1} \cdot e^{\frac{25,000 \text{ cal/gmol}}{1.9858775 \text{ cal/gmol} \cdot K} (\frac{1}{400} - \frac{1}{300}) \frac{1}{K}}$$

$$k_2 = 0.0015 \text{ min}^{-1} \cdot 35731$$

$$k_2 = 53.6 \text{ min}^{-1}$$

$$C_{A0} = \frac{P}{RT}$$

$$V_{tot} = \frac{F_{A0} \ln 10}{k_2} \frac{RT}{P}$$

$$V_{tot} = \frac{(19.1 \text{ lb mol/hr}) (2.303) (10.73 \frac{\text{ft}^3 \text{ psi}}{\text{lb mol} \cdot K}) (720R)}{53.6 \text{ min}^{-1} (\frac{60 \text{ min}}{1 \text{ hr}}) (146.7 \text{ psi})}$$

$$V_{tot} = 0.72 \text{ ft}^3$$

$$V = 0.5 \text{ in} = 0.0416667 \text{ ft}$$

$$n = \frac{V_{tot}}{V_{tube}} = \frac{0.72 \text{ ft}^3}{\pi r^2 L} = \frac{0.72 \text{ ft}^3}{\pi (0.0416667 \text{ ft})^2 (10 \text{ ft})}$$

$$n = 13.2 \text{ (round up)}$$

$$\therefore \text{Number of tubes are } 14$$